Exact Transversal Logical Groups of Quantum MDS–CSS Codes

Tavis Rudd

August 2026

Abstract

Transversal gates implement encoded operations by tensor products of single-qudit unitaries. We determine the exact projective transversal logical group for the quantum MDS–CSS codes associated with linear $[2m, m, m+1]_q$ maximum distance separable codes over odd prime fields, allowing the one-qudit gate to depend on the coordinate. The classification reduces to the code conductor

$$\mathrm{Cond}(C, C^{\perp}) = \{s \in \mathbb{F}_q^{2m} : s \star C \subseteq C^{\perp}\} = (C^{\star 2})^{\perp},$$

where $\star$ denotes coordinatewise multiplication and code products are linearly spanned. The half-rate MDS hypotheses force this space to have dimension zero or one. The exact logical group is $\mathbb{F}_q^2 \rtimes T$ in the zero-dimensional case and $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ in the one-dimensional case, where $T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^{\times}\}$ is the diagonal determinant-one subgroup, also called a split torus. Thus the codimension of the Schur square determines all transversal logical unitaries. The corresponding coordinatewise CSS endomorphism algebra is $\mathbb{F}_q \times \mathbb{F}_q$ or $M_2(\mathbb{F}_q)$. The construction is explicit, and an imported rigidity theorem for stabilizer absolutely maximally entangled states excludes non-Clifford product implementations.

For length six, the nonzero-conductor branch is equivalent to the six parity-check points lying on a conic. On a stated regular locus of an explicit non-GRS pencil, a degree-eight invariant classifies projective and monomial-code equivalence over odd fields and local-Clifford and local-unitary equivalence over odd prime fields. These geometric applications are independent of the all-length proof.

Index Terms—Absolutely maximally entangled states, Calderbank–Shor–Steane codes, conics, isodual codes, quantum MDS codes, self-association, six-arcs, transversal logical gates.

1 Introduction

A logical gate acts on encoded quantum information. A transversal implementation acts independently on the physical subsystems, so an error on one subsystem is not spread directly to another subsystem by the gate. This makes transversal gates a basic fault-tolerant primitive, but also a strongly constrained one: no quantum code admits a universal set of transversal gates [EK09]. Determining the exact transversal logical group of a code family is therefore a natural structural problem.

This paper solves that problem for a family of one-logical-qudit quantum MDS codes. Let $C \le \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ maximum distance separable (MDS) code. Its equal-phase CSS state is

$$|\Psi_C\rangle = q^{-m/2} \sum_{c \in C} |c\rangle. \tag{1.1}$$

The state is absolutely maximally entangled (AME). If one tensor factor is regarded as an input, the remaining factors define a $[[2m-1, 1, m]]_q$ quantum MDS encoder.

Throughout, a *transversal* physical unitary has the fixed-coordinate, site-dependent form

$$U_1 \otimes \cdots \otimes U_{2m-1}.$$

The factors may differ from one coordinate to another. This convention is strictly broader than the uniform form $U^{\otimes n}$. The distinction is essential here: on one branch of the classification, a logical symplectic matrix is realized at different sites through conjugations determined by coordinatewise conductor ratios. Logical groups are projective, so global scalar phases are divided out. Permutations of physical coordinates are not included unless stated explicitly.

The classification is controlled by a standard coding-theoretic object. For linear codes $E, F \leq \mathbb{F}_q^n$, their *conductor* is

$$\mathrm{Cond}(E, F) = \{s \in \mathbb{F}_q^n : s \star E \subseteq F\}.$$

We write $E \star F$ for the linear span of all coordinatewise products $e \star f$, and $C^{\star 2} = C \star C$. In the present problem,

$$\mathrm{Cond}(C, C^{\perp}) = (C^{\star 2})^{\perp}.$$

Thus the test may be computed either as a conductor or as the codimension of the Schur square. The definition of the conductor asks only for inclusion. Proposition 3.1 shows that the MDS property upgrades every nonzero inclusion to the equality $s \star C = C^{\perp}$. It also shows that the conductor has dimension zero or one. In the latter case, C is diagonally equivalent to its dual with the coordinate order fixed.

Theorem 1.1 (Exact transversal logical group)**.** Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. Then $\dim \mathrm{Cond}(C, C^{\perp})$ is zero or one, and the projective group of logical unitaries with fixed-coordinate transversal implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes T, & \dim \mathrm{Cond}(C, C^{\perp}) = 0, \\ \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \dim \mathrm{Cond}(C, C^{\perp}) = 1, \end{cases} \qquad T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^{\times}\}.$$

The second case is equivalent to $SC = C^{\perp}$ for a nonsingular diagonal matrix S.

Thus the exact transversal group is determined by one nullity computation on the classical code. In both branches, $\mathbb{F}_q^2$ acts by logical Pauli translations. The split torus T, the diagonal determinant-one subgroup, preserves the logical X- and Z-axes separately. The full $\mathrm{SL}_2(q)$ branch can mix those axes and therefore includes Fourier-like transformations exchanging logical X and Z.

The proof has three steps. First, represent each one-qudit Clifford by a 2-by-2 symplectic matrix. Preservation of the CSS stabilizer gives four block equations involving C and $C^{\perp}$. Any nonzero off-diagonal block belongs to a conductor from one code to the other. Second, the MDS distance forces every such conductor element to have full support and forces the conductor to be at most one-dimensional. If the space is zero, all local symplectic matrices are diagonal and the linear logical image is T. If it is one-dimensional, its generator constructs upper and lower unipotent matrices, which generate $\mathrm{SL}_2(q)$. Finally, the imported rigidity theorem below shows that an arbitrary product-unitary implementation is Clifford at every site, proving that the constructed groups are exact.

Prior results. The result fits into a broad study of transversal and automorphism-induced logical gates. Rains organizes coordinatewise symplectic symmetries through code-invariance algebras [Rai99, Theorems 4 and 14]. In classical coding theory, conductors are standard Schur-product objects [CMCP17, Section 2], and the product Singleton bound controls their dimensions for MDS codes [MZ15]. Compatible symplectic bases describe transversal Clifford structure for self-dual CSS codes [TTF25, Theorem 1], while uniform diagonal multiblock actions and endomorphism algebras constrain qubit stabilizer codes [DB25, Section II and Theorem 6.1]. Code automorphisms and ZX dualities give further Clifford constructions, including the Pauli corrections required to obtain the intended logical action [SKW$^+$25, Sections III–IV]. Under the more restrictive uniform convention $V_F^{\otimes n}$, Prakash and Singhal characterize odd-prime one-logical-qudit codes with a complete transversal Clifford set [PS26, Section IV.C]; Albert gives related normal forms for binary CSS codes [Alb26, Sections III–V].

The closest algebraic predecessors have different outputs and conventions. Rains's invariance algebra supplies the general container for the block preservation calculation, but does not specialize half-rate MDS–CSS label spaces or identify their fixed-coordinate logical image as exactly T or $\mathrm{SL}_2(q)$. Dasu and Burton treat diagonal transversal Clifford actions on blocks of qubit stabilizer codes and classify the associated endomorphism-algebra types; their result does not give the odd-prime, site-dependent MDS specialization here. The compatible-basis result for self-dual CSS codes gives constructions rather than the exact image for an arbitrary member of the present family. None of these cited results supplies the product-unitary exactness conclusion, which here comes from the imported stabilizer-AME rigidity theorem.

Gap and contribution. What is missing from these results is the exact fixed-coordinate, site-dependent group for the present MDS–CSS family. Theorem 1.1 supplies it by reducing the problem to $\mathrm{Cond}(C, C^{\perp})$. We give a short self-contained proof of the needed equal-parameter MDS conductor lemma, although it is also a standard consequence of product-code theory. The new step is to use this conductor to collapse the CSS invariance algebra to two types, construct both resulting logical groups, and prove exactness for all product-unitary implementations. Tan's general AME–QMDS correspondence relates state symmetries and transversal encoder gates [Tan26, Theorems 3.5–3.6]; here it transfers the state-symmetry calculation to each encoder view.

Imported inputs

The exactness argument uses four results from the companion paper: product-unitary rigidity, its encoder consequence, a stabilizer-character correction, and minimum-support transition data. Their formal statements are collected here so that the boundary between imported and proved material is explicit.

Theorem 1.2 (Imported stabilizer-AME rigidity [Rudd26a])**.** For every prime power q and $m \geq 2$, every product-unitary intertwiner between two stabilizer $\mathrm{AME}(2m, q)$ states is Clifford on each tensor factor.

Corollary 1.3 (Imported transversal Clifford no-go [Rudd26a])**.** Every tensor-product conversion between the associated stabilizer $[[2m-1, 1, m]]_q$ encoders is Clifford on every physical factor and on the logical factor. In particular, no code in this family admits a transversal non-Clifford logical unitary.

We also use one state-level consequence. A Clifford map of two stabilizer label groups need not preserve the chosen stabilizer characters, but a Pauli correction removes that discrepancy.

Lemma 1.4 (Imported stabilizer-character correction [Rudd26a])**.** For stabilizer states in the standard finite-field Weyl convention, every product-Clifford map of their stabilizer label groups can be corrected by a product Pauli to map one state ray to the other.

Theorem 1.5 (Imported minimum-support transitions [Rudd26a])**.** For a stabilizer $\mathrm{AME}(2m, q)$ state, projection from the Pauli labels supported on any $(m+1)$-coordinate set to the local Pauli-label plane at each retained coordinate is bijective. These supported subgroups span the full stabilizer label group. Under local-Clifford equivalence, the resulting transition maps transform by the corresponding local trace-symplectic changes of basis.

This is the consequence of Section 4 of the cited paper used here. We do not use its quantitative results.

Six-point applications. At length six, the coding-theoretic statement is that C is diagonally equivalent to its dual with the labels fixed. Its geometric equivalent is that the six projective columns lie on a conic, or equivalently are fixed-label Gale self-associated. Its quantum-information consequence is the enlargement of the fixed-coordinate logical symplectic image from T to $\mathrm{SL}_2(q)$. For an explicit non-GRS pencil, a degree-eight invariant z classifies projective and monomial-code equivalence over odd fields. Over odd prime fields it also classifies local-Clifford and local-unitary equivalence. A worked six-point example, deferred to an appendix, illustrates the distinction between syndrome geometry and the fixed-coordinate transversal group.

Lift, verification, and invariant refinements. The linear $\mathrm{SL}_2(q)$ action has coherent Weil lifts, whereas the affine translation subgroup has the usual Heisenberg central extension. Coordinate permutations form a separate group-extension problem. Fixed-copy scalar contractions are generically constant on regular code families, while operator-valued reduced states retain the local Weyl directions used by the rigidity theorem. The finite six-point calculations have exact certificate replays, but no finite enumeration enters the all-length theorem.

Section 2 fixes the stabilizer and MDS–CSS conventions. Section 3 proves the conductor lemma and exact group; Section 4 gives the six-point conic criterion; and Section 5 classifies the one-parameter family. Section 6 explains the limitation of fixed-copy scalar invariants. The appendices collect lift refinements, a worked syndrome example, explicit invariant calculations, transport and coordinate-permutation calculations, and the verification boundary.

2 Stabilizer AME states and MDS–CSS models

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i\ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x+a\rangle, \qquad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

After choosing dual $\mathbb{F}_p$-bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p-level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \mathrm{PCliff}(q) \longrightarrow \mathrm{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1. \tag{2.1}$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$-linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n. Its projective label group $L \le (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \tag{2.2}$$

If A is a party set, put

$$L(A) = \{\ell \in L : \operatorname{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \tag{2.3}$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

For a linear code $C \leq \mathbb{F}_q^{2m}$, the equal-phase state (1.1) has the exact stabilizer lifts

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^{2m} X(c_i)Z(h_i), \qquad (c,h) \in C \oplus C^\perp.$$

They commute and multiply without a residual phase because $h \cdot c = 0$.

Proposition 2.1 (MDS–CSS AME dictionary)**.** Let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. Then $|\Psi_C\rangle$ is a minimal-support stabilizer AME$(2m, q)$ state with projective Pauli-label Lagrangian

$$L_C = (C \oplus 0) \oplus (0 \oplus C^\perp).$$

For every m-coordinate set I, puncturing C to I is bijective, and the reduced state on I is $q^{-m}I$.

Proof. The operators $X(c)$, $c \in C$, and $Z(h)$, $h \in C^\perp$, fix the equal superposition. Their label space has dimension $2m$ and is Lagrangian. Puncturing C to any m-coordinate set is injective by the distance $m+1$, and hence bijective by dimension. In the corresponding partial trace, equality on the complementary m coordinates forces equality of the codewords, while projection to the retained coordinates runs once through $\mathbb{F}_q^m$. The reduction is therefore $q^{-m}I$; smaller reductions are maximally mixed by tracing again. The computational support has size q^m, which is minimal for an AME$(2m, q)$ state. $\square$

The rigidity and minimum-support transition theorems use only (2.1)–(2.3). For the geometric classification, we now specialize this dictionary to six coordinates.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H. The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \operatorname{rowspace} H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

Proposition 2.2 (Arc–AME dictionary)**.** For a six-arc with kernel C, the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer AME$(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Conversely, for a linear $[6,3]_q$ code C the state $|\Psi_C\rangle$ is $\mathrm{AME}(6,q)$ only if C has minimum distance four, so the dictionary is an equivalence. A monomial code equivalence induces local-Clifford equivalence of the corresponding states, with its permutation component permuting tensor factors.

Proof. The forward direction is Proposition 2.1 with $m = 3$.

For the converse, suppose the linear $[6,3]_q$ code C has a nonzero word c of weight at most three, and choose a three-party set A containing its support. Then $(c, 0)$ is a nonidentity label of L_C supported on A, so $L_C(A) \neq \{0\}$ and ρ_A is not maximally mixed. Hence an AME equal-phase state forces minimum distance four, which at length six and dimension three is the exact $[6,3,4]_q$ parameter set.

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S, let

$$L_C(S) = \{\ell \in L_C : \mathrm{supp}(\ell) \subseteq S\}.$$

The general stabilizer partial-trace formula specializes to

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \tag{2.4}$$

For later block calculations, write

$$C_X = C \oplus 0, \qquad C_Z^\perp = 0 \oplus C^\perp, \qquad L_C = C_X \oplus C_Z^\perp.$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use "GRS state" for the state of any such code. The non-GRS pencil in Section 5 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Exact MDS–CSS transversal logical groups

The classification rests on one standard MDS conductor statement. It converts every nonzero coordinatewise map between two MDS codes with the same parameters into an isomorphism and leaves room for at most one projective map.

The conductor and endomorphism-algebra statements below are valid over every finite field. Identifying their determinant-one units with the full local Clifford action uses q prime. For $q = p^e$ with $e > 1$, they describe only the $\mathbb{F}_q$-linear sector of the trace-symplectic Clifford group.

Proposition 3.1 (Equal-parameter MDS conductors)**.** Let $1 \le k < n$, and let $E, F \le \mathbb{F}_q^n$ be linear $[n, k, n-k+1]_q$ MDS codes. Every nonzero $s \in \mathrm{Cond}(E, F)$ has full coordinate support and $s \star E = F$. Consequently

$$\dim_{\mathbb{F}_q} \mathrm{Cond}(E, F) \le 1, \qquad \mathrm{Cond}(E, E) = \mathbb{F}_q(1, \ldots, 1).$$

In particular, $\mathrm{Cond}(C, C^\perp)$ has dimension zero or one, and C is diagonally equivalent to its dual exactly in the dimension-one case. When that line exists, any two nonzero witnesses differ by a unique nonzero scalar, so every ratio s_i/s_j is intrinsic.

Proof. Let $s \neq 0$ have support S of size w. Multiplication by s on E has rank $\min\{k, w\}$, the rank of puncturing an MDS code to S. Its image lies in the subcode of F supported on S, whose dimension is $\max\{0, k + w - n\}$. Hence

$$\min\{k, w\} \leq \max\{0, k + w - n\}.$$

The right side is positive. If $w < k$, the inequality would give $w \leq k + w - n$, contrary to $k < n$. Thus $w \geq k$, and the same inequality gives $w \geq n$. Therefore s has full support. Multiplication by s is invertible on the ambient space, and the inclusion $s \star E \subseteq F$ is equality by equal dimension.

If two conductor elements were independent, a nonzero linear combination could be chosen to vanish at one coordinate, contradicting full support. Hence $\dim \mathrm{Cond}(E, F) \leq 1$. The all-ones vector belongs to $\mathrm{Cond}(E, E)$, proving the self-conductor statement. Taking $F = C^{\perp}$ gives the remaining assertions. $\square$

The standard conductor identity

$$\mathrm{Cond}(E, F) = (E \star F^{\perp})^{\perp} \tag{3.1}$$

follows directly from the Euclidean inner product. Together with the product Singleton bound, it gives another proof of the dimension statement [CMCP17, MZ15]. We retained the elementary proof to make the precise MDS input visible. Specializing (3.1) gives

$$\mathrm{Cond}(C, C^{\perp}) = (C^{\star 2})^{\perp}. \tag{3.2}$$

Thus the code-to-dual test has only two outcomes: no conductor, or a unique full-support conductor line. Equivalently, the Schur square has codimension zero or one.

Fix an input coordinate j. Under the Choi correspondence, a physical product unitary implements a logical unitary L exactly when adjoining the input factor L^{-T} gives a $2m$-party product symmetry of $|\Psi_C\rangle$. We therefore classify the local symplectic blocks that preserve the state label space L_C.

Write a fixed-coordinate product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^{\perp}$ is equivalent to

$$\begin{aligned} D_\alpha C \subseteq C, \quad & D_\gamma C \subseteq C^{\perp}, \\ D_\beta C^{\perp} \subseteq C, \quad & D_\delta C^{\perp} \subseteq C^{\perp}. \end{aligned} \tag{3.3}$$

These are respectively the $X \to X$, $X \to Z$, $Z \to X$, and $Z \to Z$ preservation conditions.

To collect the four standard conductor spaces in one object, define the *algebra of coordinatewise endomorphisms*

$$\mathcal{A}_C = \begin{pmatrix} \mathrm{Cond}(C, C) & \mathrm{Cond}(C^{\perp}, C) \\ \mathrm{Cond}(C, C^{\perp}) & \mathrm{Cond}(C^{\perp}, C^{\perp}) \end{pmatrix} \subseteq M_2(\mathbb{F}_q^{2m}), \tag{3.4}$$

with coordinatewise matrix multiplication.

Proposition 3.2 (The two coordinatewise endomorphism algebras)**.** The algebra $\mathcal{A}_C$ is exactly the algebra of sitewise $\mathbb{F}_q$-linear Pauli-label maps preserving $L_C = C_X \oplus C_Z^{\perp}$, and

$$\mathrm{Cond}(C, C) = \mathrm{Cond}(C^{\perp}, C^{\perp}) = \mathbb{F}_q \mathbf{1}.$$

If $\mathrm{Cond}(C, C^{\perp}) = 0$, then $\mathcal{A}_C \cong \mathbb{F}_q \times \mathbb{F}_q$. Otherwise, after choosing $s \star C = C^{\perp}$, there is an algebra isomorphism

$$M_2(\mathbb{F}_q) \longrightarrow \mathcal{A}_C, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a\mathbf{1} & bs^{-1} \\ cs & d\mathbf{1} \end{pmatrix}.$$

The coordinatewise determinant-one units are respectively T and $\mathrm{SL}_2(q)$.

Proof. The four entries in (3.4) are precisely the four inclusions in (3.3). They are closed under coordinatewise matrix multiplication: for example,

$$\mathrm{Cond}(C^{\perp}, C) \star \mathrm{Cond}(C, C^{\perp}) \subseteq \mathrm{Cond}(C, C),$$

because multiplication first sends C into $C^{\perp}$ and then back into C. The other products are identical. The conductor identity (3.1) gives

$$\mathrm{Cond}(C, C) = (C \star C^{\perp})^{\perp} = \mathrm{Cond}(C^{\perp}, C^{\perp}),$$

and Proposition 3.1 makes this common algebra the scalar line. If $\mathrm{Cond}(C, C^{\perp}) = \mathbb{F}_q s$ is nonzero, then s has full support and $\mathrm{Cond}(C^{\perp}, C) = \mathbb{F}_q s^{-1}$; the converse follows in the same way. Thus the two off-diagonal conductors either both vanish or are these two inverse lines. This gives the two displayed algebra types. In the matrix-algebra case the determinant at every coordinate is $ad - bc$; in the split case it is ad. Imposing determinant one proves the final statement. $\square$

The proposition packages the block calculation into the split-algebra versus matrix-algebra dichotomy: the MDS property leaves no intermediate coordinatewise linear symmetry algebra. We now use its two cases to prove the logical-group theorem.

Proof of Theorem 1.1. Zero-conductor case. Proposition 3.2 identifies every label-preserving symplectic action with an element of T, realized by the same block $\mathrm{diag}(a, a^{-1})$ at every coordinate. Conversely, these blocks preserve L_C. Logical Paulis have physical Pauli representatives, so this branch gives $\mathbb{F}_q^2 \rtimes T$.

Nonzero-conductor case. Now suppose

$$S = \mathrm{diag}(s_1, \ldots, s_{2m}), \qquad SC = C^{\perp}. \tag{3.5}$$

The projective vector $[s_1 : \cdots : s_{2m}]$, and hence every propagation ratio s_i/s_j, depends only on C.

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^{-}(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \qquad F_i^{+}(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \tag{3.6}$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^{\perp})$, the product of the lower blocks sends

$$(c, h) \longmapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \longmapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries: Lemma 1.4 supplies the required product-Pauli corrections. Thus label-space preservation is sufficient here; no coherent choice of Clifford phases is being assumed. At any chosen input party j, their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\mathrm{SL}_2(q)$. Under the Choi correspondence, an input unitary U implements the logical unitary $U^{-T} = \overline{U}$. Complex conjugation fixes $X(a)$ and sends $Z(b)$ to $Z(-b)$, so an input block A induces the logical block KAK with $K = \mathrm{diag}(1, -1)$. This map is an automorphism of $\mathrm{SL}_2(q)$ preserving the diagonal torus, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ is transversal.

Exactness. The imported Corollary 1.3 excludes every product-unitary logical action outside that group. Hence the displayed group is exact. $\square$

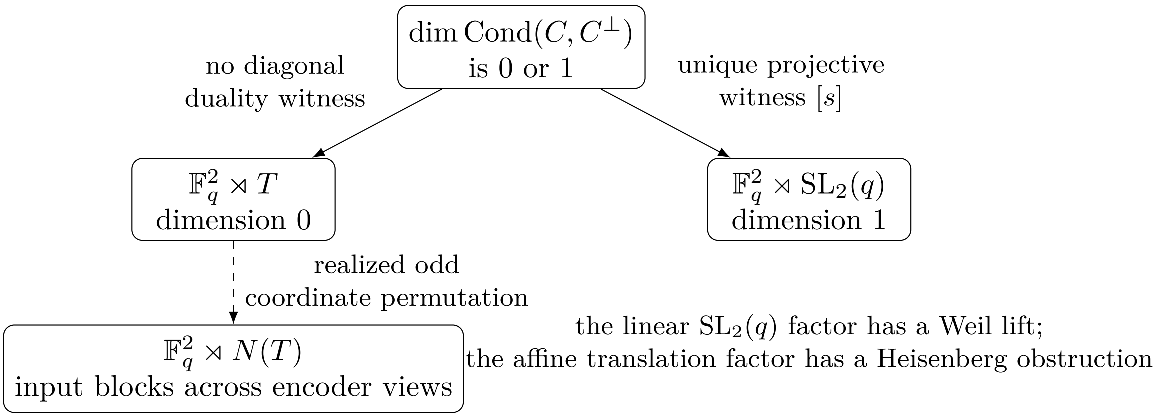

Figure 1: The dimension of the code-to-dual conductor selects the exact fixed-coordinate projective transversal group. Coordinate permutations form a separate extension problem: in the non-GRS examples, an odd permutation enlarges the possible input blocks from T to $N(T)$ across encoder views, where $N(T)$ is the normalizer of T in $\mathrm{SL}_2(q)$; the permutation need not fix the chosen encoder input.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (3.5), including the extended evaluation point at length $q+1$ [HP03, Sections 5.2–5.3]. Thus the GRS tower lies on the branch diagonally equivalent to its dual, although the proof uses no evaluation structure.

Proposition 3.3 (Site-dependent propagation)**.** Fix an input coordinate j, put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is the twisted diagonal embedding

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1} b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \tag{3.7}$$

Proof. The formula follows from the upper and lower elementary matrices in (3.6), which generate $\mathrm{SL}_2(q)$. $\square$

The formula is the reason for allowing the one-qudit factor to depend on the coordinate: a common logical matrix generally propagates differently at different sites. Appendix A asks when fixed local frame changes can make these actions uniform and records two phase refinements. The linear $\mathrm{SL}_2(q)$ factor admits coherent Weil lifts, whereas adjoining logical Pauli translations gives the nonsplit Heisenberg central extension.

4 The six-point conic criterion

Section 3 already determines the fixed-coordinate transversal logical group at every length. In the six-coordinate case, the same coding-theoretic test has a classical geometric interpretation: the conductor $\mathrm{Cond}(C, C^{\perp})$ is nonzero exactly when the six projective points lie on a conic.

Throughout this section q is an odd prime. Regard any one coordinate of an $\mathrm{AME}(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, a one-qudit Clifford acts symplectically through $\mathrm{SL}_2(q)$, so the fixed-coordinate symplectic stabilizer of the label space is

$$\mathcal{K}_C = \{(F_1, \dots, F_6) \in \mathrm{SL}_2(q)^6 : (\bigoplus_i F_i)L_C = L_C\}.$$

For an encoder view, the *fixed-coordinate logical symplectic image* is the image of $\mathcal{K}_C$ on its input block.

Theorem 1.1 expresses the dichotomy as diagonal equivalence to the dual. In projective geometry, the same condition is called fixed-label Gale self-association. The labeled Gale transform consists of the six columns of any generator matrix for $\ker H$. Fixed-label self-association means that this configuration is projectively equivalent to the columns of H without permuting their labels. The next lemma identifies this condition with the conic locus.

Lemma 4.1 (Fixed-label self-association)**.** Let H be a rank-three 3×6 matrix whose columns are a six-arc, and put $C = \ker H$. There is a nonsingular diagonal matrix S with $SC = C^{\perp}$ if and only if the six columns of H lie on a conic.

Proof. Let G be a 3×6 generator matrix for C. Its columns are the labeled Gale transform of the columns of H. Since $C^{\perp} = \mathrm{row}(H)$,

$$SC = C^{\perp} \iff \mathrm{row}(GS) = \mathrm{row}(H) \iff AGS = H \text{ for some } A \in \mathrm{GL}_3(q).$$

Thus the condition is precisely fixed-label projective equivalence to the Gale transform. This fixed locus can be identified directly over the finite field. Put

$$V_i = (x_i^2, y_i^2, z_i^2, x_i y_i, x_i z_i, y_i z_i)$$

and let V be the 6-by-6 matrix with rows V_i. The six entries are the degree-two monomials in three variables, equivalently the independent coefficients of a symmetric 3-by-3 matrix. Thus a vector in the kernel of V is a quadratic form vanishing on all six points, and $\det V = 0$ is exactly the conic condition.

Any five arc points impose independent conditions on conics. Indeed, a relation among at most five rows of V would give $H_I D H_I^T = 0$ for a nonsingular diagonal D on its support I. For $|I| \geq 3$, this would put the three-dimensional row space of $H_I D$ inside $\ker H_I$, which has dimension at most two; smaller supports are immediate. Hence any relation $\sum_i d_i V_i = 0$ has every $d_i \neq 0$.

Writing $D = \mathrm{diag}(d_i)$, that relation is $HDH^T = 0$. Therefore $\mathrm{row}(HD) \subseteq \ker H = C$, and equality follows from dimension three. Thus $CD^{-1} = C^{\perp}$. Conversely, a nonsingular diagonal equivalence $SC = C^{\perp}$ gives $HS^{-1}H^T = 0$, hence a relation among the rows of V and a conic through the six columns. The conic is irreducible because the columns form an arc. This is the finite-field fixed-label form of the classical six-point self-association theorem [HMSV08, Section 2.3]. $\square$

Theorem 4.2 (Six-point transversal-group criterion)**.** For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-coordinate logical symplectic image is

$$\begin{cases} \mathrm{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^{\times}\}, & \text{if the arc is nonconic.} \end{cases} \tag{4.1}$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-coordinate symmetries. Off the GRS locus, the image is the split torus.

Proof. Theorem 1.1 gives $\mathrm{SL}_2(q)$ exactly when C is diagonally equivalent to its dual, and T otherwise. Lemma 4.1 identifies this condition with the conic locus. □

Adding logical Paulis recovers the exact fixed-coordinate projective logical groups in Theorem 1.1. This is an operational distinction, not a classical automorphism count: the non-GRS group has order $q^2(q-1)$, whereas the GRS group has order $q^2|\mathrm{SL}_2(q)|$.

Allowing coordinate permutations is a separate problem. A permutation that exchanges the CSS X- and Z-spaces can supply the nontrivial Weyl-group element normalizing T, but acts on a fixed encoder view only when it fixes the input coordinate; Appendix F treats this extension. Appendix C gives an optional worked example separating syndrome symmetry from the parity-check conic criterion.

5 A non-GRS pencil and its quotient

We next give a substantial six-point application. The family below contains both conic and nonconic members and retains nontrivial projective variation away from the conic divisor. It is useful because its projective equivalence problem has an explicit one-dimensional quotient, so the geometric branch of the all-length theorem can be seen in a complete family rather than at an isolated code.

Over a finite field of odd order, put

$$H_t = \big((0,1,1-t),(0,1,t-1),(1,1-t,0),(1,t-1,0),(1,0,-t),(1,0,t)\big),$$

and define

$$A(t) = -4t(t-1)^2, \qquad B(t) = (t^2-t+1)(t^2-3t+1), \qquad z(t) = \big(B(t)/A(t)\big)^2.$$

The form of z is dictated by the residual parameter symmetries. The intermediate coordinate

$$y(t) = \frac{(t-1)^2}{t}$$

is generically two-to-one in t, and projective equivalence identifies y with $\pm y^{\pm 1}$. Since $z = (y-y^{-1})^2/16$, the map $t \mapsto z$ is generically eight-to-one.

The admissibility conditions have separate meanings. The factors $t(t-1)(t^2-t+1)(t^2-3t+1)$ exclude arc degenerations and singularities of the displayed quotient coordinates. The factor

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1$$

cuts out the conic, equivalently GRS, locus. Define the regular non-GRS set

$$\mathcal{U} = \{t : t(t-1)(t^2-t+1)(t^2-3t+1)G(t) \neq 0\}. \tag{5.1}$$

Factoring the twenty 3-by-3 column minors gives the first four factors, while the determinant for a conic through the six columns is a unit times G. For $t \in \mathcal{U}$, write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$.

Here $\sim_{\mathrm{proj}}$ means projective equivalence of the six-point configurations after independent column rescaling and coordinate permutation; $\sim_{\mathrm{mon}}$ is monomial code equivalence; and $\sim_{\mathrm{LC}}$, $\sim_{\mathrm{LU}}$ allow tensor-factor permutations followed by products of one-qudit Clifford or unitary operators, respectively.

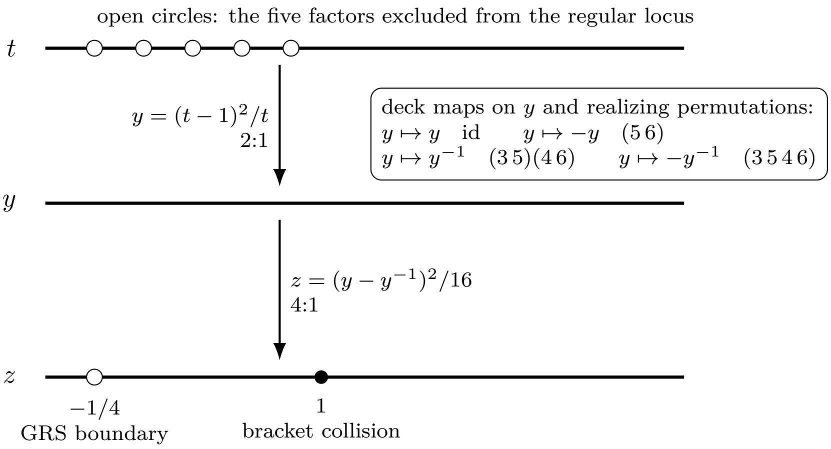

Figure 2: The degree-eight quotient of the regular non-GRS pencil (special points on the z-line are arranged schematically). The parameter t double-covers $y = (t-1)^2/t$, and the four identifications $y \mapsto \pm y^{\pm 1}$, each realized by a displayed coordinate permutation, cover z. The open point $z = -1/4$ is the excluded GRS boundary; $z = 1$ is the bracket collision handled in the proof.

Theorem 5.1 (Projective and Clifford quotient)**.** For $t, u \in \mathcal{U}$, the following are equivalent:

$$H_t \sim_{\text{proj}} H_u, \qquad C_t \sim_{\text{mon}} C_u, \qquad z(t) = z(u),$$

allowing arbitrary relabelings of the six coordinates. If q is prime, these are also equivalent to $\Psi_t \sim_{\text{LC}} \Psi_u$.

Corollary 5.2 (Quantum classification on the regular locus)**.** If q is an odd prime and $t, u \in \mathcal{U}$, then

$$H_t \sim_{\text{proj}} H_u \iff C_t \sim_{\text{mon}} C_u \iff \Psi_t \sim_{\text{LC}} \Psi_u \iff \Psi_t \sim_{\text{LU}} \Psi_u \iff z(t) = z(u),$$

where the code and point relabeling is the corresponding tensor-factor permutation for the quantum states.

Proof of Theorem 5.1. The proof has two independent recovery steps. First, equality of z is equivalent to the residual orbit relation $y(u) \in \{\pm y(t)^{\pm 1}\}$. Each of these four relations is realized by an explicit projectivity and coordinate permutation. For the converse, write $[ijk]$ for the determinant of columns i, j, k. The ten signed products $[ijk][\ell mn]$ over complementary triples have multiplicity pattern

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\},$$

which recovers $(B/A)^2 = z$, including the collision $z = 1$. This proves the projective classification and, through the MDS dictionary of Section 2, the monomial-code classification. These claims are Proposition B.1, whose proof in Appendix B lists all four projectivities and checks the bracket invariant.

For prime q, Theorem 1.5 assigns a two-dimensional stabilizer-label space to every four-coordinate support. Projection identifies that space with the Pauli-label plane at each retained coordinate. Composing one such identification with the inverse of another gives a transition map, and closed chains of transitions give endomorphisms M of a local Pauli-label plane. Their projective conjugacy invariant is

$$r(M) = \operatorname{Tr}(M)^2 / \det M,$$

which is unchanged by conjugation, scalar multiplication, and inversion. Its multiplicity data recover $-1/z$, so local-Clifford equivalence also forces equality of z. This is Proposition B.2; its proof in Appendix B defines the transition maps, and Lemma D.1 reduces the multiplicity calculation to ten paired-coordinate orbits. Projective equivalence directly supplies a local-Clifford equivalence. □

Proof of Corollary 5.2. Theorem 5.1 gives all equivalences except local-unitary equivalence. Every local Clifford is local unitary, while Theorem 1.2 turns every product-unitary intertwiner between Ψ_t and Ψ_u into a product-Clifford intertwiner. □

The projective quotient remains valid over odd extension fields, but the local-Clifford and local-unitary conclusion is intentionally restricted to prime fields. Frobenius maps are already additional local Clifford operations, and $z(t^p) = z(t)^p$ need not equal $z(t)$. Appendix B records the Frobenius-sector divisors and a concrete $\mathbb{F}_{25}$ example; no complete extension-field orbit classification is claimed.

6 Limits of fixed-copy scalar invariants

The operator-valued marginal mechanism retains local Weyl axes. Fixed-copy scalar contractions retain less: on algebraic families they are generically constant and can distinguish parameters only on rank-jump strata. We make that limitation precise here; Appendix D records two explicit separators on exceptional strata.

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a d-dimensional code, let $M_{\boldsymbol{\sigma}}(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \le i \le n,\ 1 \le a \le r).$$

Counting its solutions gives the normalized contraction

$$I_{\boldsymbol{\sigma}}(\Psi_C) = q^{dr - \operatorname{rank} M_{\boldsymbol{\sigma}}(G)}. \tag{6.1}$$

The next statement uses a small amount of algebraic-geometric language. An irreducible parameter variety is a family not decomposed into smaller closed families, and a dense open subset is what remains after excluding the common zero locus of finitely many nonzero polynomial conditions. Thus "generically constant" means constant away from such rank-jump conditions.

Proposition 6.1 (Generic constancy of fixed-copy invariants)**.** Fix r and a field k of cardinality q, which is also the local Hilbert-space dimension under the code–state dictionary. Fix an irreducible reduced parameter variety X over k, and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree-(r,r) permutation contraction of the associated equal-phase states has one value at every k-point of some dense open subscheme $U \subseteq X$ defined over k. The same holds for party-orbit sums and linear combinations. After evaluating the states as vectors in $(\mathbb{C}^q)^{\otimes n}$, in the stable range $q \ge r$, every scalar LU-invariant of bidegree (r,r) is generically constant in this sense. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.1) depends only on the rank of a matrix whose entries are regular functions over k. A maximal nonzero minor shows that this rank is constant on a dense open subscheme defined over k. There are only finitely many diagrams at fixed r, so their generic-rank opens have a common dense intersection because X is irreducible. This proves the geometric generic-rank assertion. Over the complex vector space $(\mathbb{C}^q)^{\otimes n}$, applied pointwise to each code state, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree-(r, r) invariants; when the local Hilbert-space dimension q is at least r, there are no additional dimension relations among these diagrams [CŚ06, Section 2]. □

Remark 6.2 (Finite-field points). The dense open subscheme U need not contain a k-rational point. If $U(k) = \varnothing$, the corresponding pointwise statement over the finite field is vacuous; the geometric generic-rank statement remains valid.

Thus no fixed copy number recovers a nonconstant pencil coordinate as a regular function on the generic locus. This is a statement about the geometric generic locus, not about the distinguishability of individual code states: polynomial LU invariants separate inequivalent states, so any finite collection of pairwise inequivalent code states is separated at some finite copy number. What remains open is whether that copy number can be bounded uniformly as q or the family grows. A scalar witness can still detect a special rank-jump stratum, as the four-copy example in Appendix D does. In contrast, the four-party reduced-operator family (2.4) retains the Weyl axes, and the imported rigidity theorem forces Cliffordness. This is the precise distinction between generic constancy of scalar invariants and the operator-valued marginal information used by the rigidity theorem.

7 Conclusion

The code conductor turns the transversal-group problem into a one-dimensional linear test. For the present half-rate MDS code, $\mathrm{Cond}(C, C^{\perp}) = (C^{\star 2})^{\perp}$ is either zero or a full-support line. Equivalently, the coordinatewise endomorphism algebra of the CSS label space is either $\mathbb{F}_q \times \mathbb{F}_q$ or $M_2(\mathbb{F}_q)$. Their determinant-one units give the split torus and $\mathrm{SL}_2(q)$, respectively. Stabilizer-AME rigidity then proves that these Clifford constructions exhaust every product-unitary logical implementation.

For six coordinates, the same dichotomy becomes the conic criterion in the six-arc model. Further geometric and symmetry refinements show how the all-length criterion organizes finer finite geometry without depending on those calculations. Fixed-copy scalar contractions are generically constant, whereas the operator-valued reduced states retain the local Weyl directions. This explains why the rigidity argument succeeds where bounded-degree scalar invariants do not.

Several questions remain. Over extension fields, the full additive Clifford group can contain nonsemilinear sectors not detected by the prime-field statement. For coordinate-permuting symmetries, the nonabelian factor set gives an exact splitting criterion, and the integral H_3 family now has a uniform complement; a conceptual classification beyond that family is open. Beyond MDS codes, the same coordinatewise algebra may acquire a square-zero off-diagonal radical, suggesting Borel subgroups as the first intermediate field-linear CSS symmetry type. A general theorem must also account for decomposable codes, positive logical dimension, and prime-power trace-symplectic effects, and is left for future work.

A Physical realizations and lift refinements

This appendix refines the projective classification of Section 3. None of its results is needed to identify the exact projective logical group.

A.1 Uniform local frames

Assume that $S = \operatorname{diag}(s_1, \ldots, s_{2m})$ satisfies $SC = C^{\perp}$. Fix an input coordinate j, put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

The site-dependent realization from Proposition 3.3 is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1}b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}.$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (3.6) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. The calculation applies at the extended evaluation point and uses only diagonal equivalence $SC = C^{\perp}$, not the evaluation formula for a GRS code.

The following refinement is not needed for the transversal-group classification; it answers whether the site-dependent actions can be made identical by changing the local Clifford frame at each coordinate.

Proposition A.1 (Uniform local frames)**.** The realization in Proposition 3.3 can be made uniform by fixed local Clifford frame changes if and only if every ratio $t_i = s_i/s_j$ is a square. Equivalently, the obstruction is the collection of relative determinant square classes in

$$\mathrm{PGL}_2(q)/\mathrm{PSL}_2(q) \cong \mathbb{F}_q^{\times}/(\mathbb{F}_q^{\times})^2.$$

Proof. Conjugation by $\operatorname{diag}(1, t_i)$ is inner on $\mathrm{SL}_2(q)$ exactly when t_i is a square. Indeed, the centralizer of $\mathrm{SL}_2(q)$ in $\mathrm{GL}_2(q)$ is scalar, and a scalar multiple of this diagonal matrix has determinant one exactly in the square case. Since $t_j = 1$, a nonsquare t_i prevents uniformization by fixed local Clifford frame changes: the i-th factor represents the diagonal outer automorphism. Equivalently, the obstruction is the relative class of t_i in

$$\mathrm{PGL}_2(q)/\mathrm{PSL}_2(q) \cong \mathbb{F}_q^{\times}/(\mathbb{F}_q^{\times})^2.$$

Here $\mathrm{PGL}_2(q) = \mathrm{GL}_2(q)/\mathbb{F}_q^{\times}$, and $\mathrm{PSL}_2(q)$ is the image of $\mathrm{SL}_2(q)$ in that quotient. □

A.2 Coherent lifts and the affine obstruction

The projective classification records the action on Pauli labels. We next ask whether unitary representatives can be phased coherently so that they respect multiplication exactly.

For odd prime q, and more generally for every odd prime power, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1].[1] Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A. In the symmetric Weyl convention $D_{a,b} = \chi(ab/2)X(a)Z(b)$, a CSS label $(c, h) \in C \oplus C^{\perp}$ has tensor lift

$$\bigotimes_i D_{c_i,h_i} = \chi(c \cdot h/2)X(c)Z(h) = X(c)Z(h),$$

because $c \cdot h = 0$. Exact Egorov covariance therefore preserves the CSS stabilizer group with its trivial character, and hence the state ray, rather than only its projective label space. Consequently the one-qudit Clifford scalar extension restricted to the linear $\mathrm{SL}_2(q)$ factor splits coherently with the propagated tensor symmetries.

[1]The exceptional uniqueness clause at $q = 3$ in Gérardin's theorem does not affect existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \qquad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent lift is therefore $H_q \rtimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Remark A.2 (Coordinate permutations). Neither lift question decides what happens when physical coordinates may be permuted. That is a separate extension of the fixed-coordinate projective group by the realized coordinate-permutation group. Appendix F states its splitting criterion, proves a uniform icosahedral family, and points to supplementary finite examples.

B Pencil quotient calculations and extension fields

This appendix supplies the explicit calculations used in Theorem 5.1. The theorem's conceptual reduction appears in Section 5; the formulas below make that reduction auditable.

B.1 The projective quotient

Proposition B.1 (Residual orbit and bracket recovery)**.** For $t, u \in \mathcal{U}$, equality $z(t) = z(u)$ holds exactly when $y(u) \in \{\pm y(t)^{\pm 1}\}$. Every such relation is realized by a projectivity and a coordinate permutation, while the complementary-bracket multiset of H_t determines $z(t)$.

Proof. With $y = (t-1)^2/t$, one has $A = -4t^2 y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \tag{B.1}$$

Each relation is realized by a projectivity and a coordinate permutation:

relation	projectivity
$y(u) = y(t)$	$\mathrm{diag}(1, (1-u)/(1-t), u/t)$
$y(u) = -y(t)$	$\mathrm{diag}(1, (1-u)/(1-t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$.

The corresponding permutations are the identity, $(5\,6)$, $(3\,5)(4\,6)$, and $(3\,5\,4\,6)$. All denominators and determinants are nonzero on $\mathcal{U}$.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \tag{B.2}$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; a coordinate permutation changes at most their common sign. On $\mathcal{U}$, $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two distinguish the two opposite pairs and recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset collapses to two opposite values of multiplicity five, which is equivalent to $z = 1$. Thus the multiset recovers z in every case. $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (B.1), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit coordinate permutations reverse w, and their projectivities lift to monomial local Cliffords.

B.2 Recovery from minimum-support transitions

Proposition B.2 (Holonomy recovery)**.** For prime q, the projective conjugacy invariants of the minimum-support transition loops of Ψ_t, counted with multiplicity, determine $z(t)$ on $\mathcal{U}$.

Proof. Specialize the imported minimum-support transition theorem to $m = 3$. It gives a canonical atlas of two-dimensional stabilizer planes, one for each four-coordinate support. Each plane identifies with the local Pauli-label plane at any coordinate in its support, and the supported planes generate the full stabilizer label group. Overlaps therefore give coordinate changes between local Weyl frames; compositions give holonomy data intrinsic up to local symplectic conjugacy.

For prime q, put $K_T = L_{C_t}(T)$ for every four-set T. If $i, j \in T$, projection $p_i : K_T \to \mathbb{F}_q^2$ is an isomorphism. Define

$$M_T(i,j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j, define

$$\mathrm{Hol}(T,U;i,j) = M_U(j,i) M_T(i,j). \tag{B.3}$$

A local Clifford sends this endomorphism to a conjugate. Reversing T, U takes its inverse, and a coordinate permutation only relabels the data. Hence the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is invariant under local Cliffords and coordinate permutations: r is unchanged by conjugation, scalar multiplication, and inversion.

Lemma D.1 derives the complete spectrum from ten orbits of the natural paired-coordinate symmetry group. In particular, one family has value $-1/z$, and its multiplicity identifies that value canonically even when specialization makes families collide. The holonomy data therefore recover $-1/z$. $\square$

B.3 Extension-field caveat

The projective quotient remains valid over odd extension fields, but the prime-field local-Clifford and local-unitary statement does not: Frobenius maps are additional local Clifford operations. The next proposition records the scalar conditions for two Frobenius sectors without asserting a complete extension-field classification.

Proposition B.3 (Frobenius-sector divisors)**.** Let σ be an automorphism of a finite field of odd order and let $t \in \mathcal{U}$. Put

$$E_\sigma(t) = (\sigma(t) - t)(1 - \sigma(t)t), \qquad D_\sigma(t) = ((1 - \sigma(t))(1 - t))^2 + \sigma(t)t.$$

Then

$$\frac{\sigma(t)}{(1 - \sigma(t))^2} = \frac{t}{(1-t)^2} \iff E_\sigma(t) = 0. \tag{B.4}$$

If $h = (1 - \sigma(t))(1 - t)$ and $w = (-h, -h, 1, 1, -1, -1)$, then

$$H_{\sigma(t)} \operatorname{diag}(w) H_t^{\mathsf{T}} = 0 \iff D_\sigma(t) = 0. \tag{B.5}$$

Also $D_{\mathrm{id}}(t) = G(t)$, and
$$E_\sigma(t) = 0 \quad \Longrightarrow \quad D_\sigma(t) \neq 0. \tag{B.6}$$
Finally G, A, B, z commute with σ, and σ preserves the regular non-GRS locus $\mathcal{U}$.

Proof. Membership in $\mathcal{U}$ gives $t \neq 1$, and injectivity of σ gives $\sigma(t) \neq 1$. Cross multiplication in (B.4) gives
$$s(1-t)^2 - t(1-s)^2 = (s-t)(1-st)$$
with $s = \sigma(t)$. Direct multiplication in (B.5) makes eight entries vanish identically and leaves $-2D_\sigma(t)$ in the last entry; odd characteristic gives the equivalence. Setting σ to the identity yields $D_{\mathrm{id}} = G$.

If $E_\sigma(t) = 0$, then either $\sigma(t) = t$ or $\sigma(t)t = 1$. In the first case $D_\sigma(t) = G(t)$. In the second, multiplying $D_\sigma(t)$ by t^2 gives $G(t)$. Membership in $\mathcal{U}$ excludes $t = 0$ and $G(t) = 0$, proving (B.6). The equivariance statements follow by applying σ to the defining polynomial and rational expressions. $\square$

Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p}, while $z(t^p) = z(t)^p$ need not equal $z(t)$. For example, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2-2)$, both s and $-s = s^5$ lie in $\mathcal{U}$, but
$$z(s) = 3 + 3s, \qquad z(-s) = 3 + 2s.$$
Proposition B.3 gives the collision and diagonal-equivalence divisors relating t to $\sigma(t)$. It does not show that an arbitrary additive local Clifford determines a field automorphism or construct a local Clifford from a Galois match of z. Neither a complete extension-field local-Clifford nor local-unitary classification is claimed.

C Syndrome geometry of a six-point example

This appendix changes from transversal gates to translated-state error geometry. Its conic lies in an error-syndrome plane and must not be confused with the parity-check conic of Section 4. The example occurs in the pencil of Section 5: over $\mathbb{F}_{11}$, the parameter $t = 2$ has $z = 1$ and gives the code called the Clebsch or H_3 code in [Rudd26b]. The following lemma supplies the coding-theoretic dictionary.

Lemma C.1 (Coset states and three-party syndrome charts)**.** Let H be a rank-three parity-check matrix whose six columns form an arc, put $C = \ker H$, and set
$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \qquad |\Psi_C\rangle = q^{-3/2} \sum_{c \in C} |c\rangle.$$
Then $|\Psi_e\rangle = |\Psi_f\rangle$ if and only if $e - f \in C$, and states from distinct cosets are orthogonal. In the row basis of H, the $Z(C^\perp)$-stabilizer syndrome of $|\Psi_e\rangle$ is He^T. Consequently the least support of an X-operator producing $|\Psi_e\rangle$ is the minimum Hamming weight in $e + C$.

For every syndrome $s \in \mathbb{F}_q^3$ and every three-party set I, there is a unique representative e supported in I with $He^\mathsf{T} = s$. If the coset with syndrome s has minimum weight three, this representative has support exactly I.

Proof. Translation replaces the computational support C by the affine coset $e + C$. Two such supports are equal precisely when $e - f \in C$; otherwise they are disjoint, which also proves orthogonality. If h is a row of H, then $h \in C^\perp$, and $Z(h)$ acts on the entire coset $e + C$ by the character $\chi(h \cdot e)$. Reading this for the three rows of H gives He^T.

Now fix I. The arc condition says exactly that the three columns in I form an invertible matrix H_I. Thus $H_I e_I^\mathsf{T} = s$ has one solution, and extension by zero gives the unique representative supported in I. If one of its three coordinates vanished, the same syndrome would have a representative of weight at most two. A minimum weight-three syndrome therefore uses all three coordinates. ☐

Proposition C.2 (Extremal X-syndrome conic)**.** Let H be a parity-check matrix whose columns are the Clebsch six-arc in $\mathbb{P}^2(\mathbb{F}_{11})$, put $C = \ker H$, and write

$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \qquad |\Psi_C\rangle = 11^{-3/2} \sum_{c \in C} |c\rangle.$$

The syndromes whose cosets have minimum X-support three are the 120 nonzero vectors on twelve projective rays forming a nonsingular conic in $\mathbb{P}^2(\mathbb{F}_{11})$. Each such syndrome has exactly one weight-three representative on each of the $\binom{6}{3} = 20$ three-party supports. The monomial automorphism group $C_{10} \times A_5$ is transitive on the 120 syndromes: A_5 is transitive on the twelve rays with point stabilizer C_5, and C_{10} is transitive on the ten nonzero vectors of each ray. Nevertheless, the defining six-arc is nonconic, so the fixed-coordinate logical symplectic image of every encoder view is the split torus T, not $\mathrm{SL}_2(11)$.

Proof. Lemma C.1 identifies minimum X-support with coset-leader weight and gives one representative on every three-party support. The Clebsch deep-hole theorem identifies the covering radius as three and the twelve projective maximum-distance syndromes with a nonsingular conic; it also gives the transitive $C_{10} \times A_5$ action [Rudd26b, Proposition 6.2, Corollary 6.3, and Proposition 6.4]. Finally, the defining arc is non-GRS and hence nonconic. The last assertion is therefore Theorem 4.2 at $q = 11$. ☐

Remark C.3 (Error symmetry without fixed-coordinate duality). The one-per-support statement means that the same extremal translated state can be created by a minimum X-operator on either side of every balanced $3 \mid 3$ cut. The conic belongs to the X-error syndrome plane; it does not put the six parity-check columns on a conic. Thus a highly symmetric error shell need not supply a Fourier block exchanging X and Z in the fixed-coordinate transversal group.

Changing the row basis of H changes the displayed conic by a projective coordinate transformation. The covering radius concerns only generalized-X translates of the six-leg AME state; it does not say that the associated $[[5, 1, 3]]_{11}$ output code corrects arbitrary weight-three errors. Theorem 1.2 supplies the separate exact rigidity statement: every product-unitary intertwiner from this tensor to a stabilizer AME tensor is Clifford factor by factor.

Remark C.4 (Coordinate permutations). For this icosahedral family, Theorem F.1 identifies the larger realized coordinate image as S_5. Odd coordinate motion inverts the fixed split torus T. This party-moving symmetry is separate from the fixed-coordinate transversal group discussed above.

A different MDS uniformity governs operator pushing. Fix an input coordinate and a nonidentity Pauli label. Each of the ten four-sets containing that coordinate carries a unique stabilizer representative with that support; removing the input leaves a weight-three output Pauli. Thus a fixed input Pauli has ten minimum-support pushes. Lemma C.1 instead fixes an X-syndrome and varies over all twenty three-coordinate supports. The comparison separates two uses of MDS uniformity and explains why symmetry of the error-syndrome shell does not enlarge the fixed-coordinate logical group.

D Explicit invariant calculations

This appendix derives the orbit formula for the minimum-support holonomy invariant and gives two explicit scalar LU separators. The holonomy calculation supports the pencil classification. The scalar separators are not used in the rigidity or logical-group results; they show what finite-copy contractions can detect on exceptional strata despite the generic-constancy result in Section 6.

D.1 The holonomy orbit formula

Let

$$\mathcal{P} = \{\{1,2\},\{3,4\},\{5,6\}\}, \qquad W = \mathrm{Stab}_{S_6}(\mathcal{P}) \cong C_2 \wr S_3.$$

The three pairs in $\mathcal{P}$ are visible in the displayed columns of H_t. They reduce the apparent 450-case calculation to ten orbit representatives.

Lemma D.1 (Holonomy orbit formula)**.** On the regular pencil locus, the minimum-support holonomy multiset is

$$\{4^{[90]},\ (-1/z)^{[144]},\ ((2z+1)^2/z^2)^{[24]},\ \lambda_+^{[96]}, \lambda_-^{[96]}\}, \tag{D.1}$$

where multiplicity is written in brackets and $\lambda_\pm$ are the two roots, over an algebraic closure, of

$$X^2 - 8X + (8 - 16z - 1/z).$$

Proof. An unoriented holonomy loop is indexed by a pair of coordinates e and an unordered pair of distinct four-supports containing e. Equivalently, write it as $(ij; ab, cd)$, where $ij = e$ and ab, cd are the two omitted pairs, each disjoint from e. There are $15\binom{6}{2} = 225$ such loops. The group W has ten orbits on them. Subscripts below give their orbit sizes:

$$\begin{aligned}
4 :&\quad (12;56,34)_3,\ (12;46,35)_6,\ (13;46,25)_{24},\ (13;24,56)_{12};\\
-1/z :&\quad (12;56,36)_{24},\ (13;46,45)_{24},\ (13;45,25)_{24};\\
(2z+1)^2/z^2 :&\quad (12;46,45)_{12};\\
\lambda_+ :&\quad (13;56,25)_{48};\\
\lambda_- :&\quad (13;24,46)_{48}.
\end{aligned}$$

This orbit decomposition is purely combinatorial. The incidences of ij, ab, cd with the three blocks of $\mathcal{P}$ distinguish the ten representatives, and orbit–stabilizer gives the displayed sizes; their sum is 225.

Substitution of the shortened X- and Z-generators into (B.3) shows first, on generators of W, that the rational function $r(M)$ is constant on each displayed orbit. Evaluating one representative of each orbit then gives the displayed values. For the final two representatives, their sum is 8 and their product is $8 - 16z - 1/z$, which gives the stated quadratic. Reversing the two four-supports replaces the holonomy by its inverse and does not change $r(M) = \mathrm{Tr}(M)^2/\det M$. Each unoriented orbit is therefore counted twice in the ordered $15 \cdot 6 \cdot 5 = 450$ multiset. The orbit sizes give $90, 144, 24, 96, 96$, proving (D.1). $\square$

The paper-local certificate independently replays the ten representative substitutions and reconstructs the same multiset directly from the CSS Lagrangian.

Because $A, B \neq 0$ on $\mathcal{U}$, we have $z \neq 0$. The two quadratic roots in (D.1) are taken over an algebraic closure; their symmetric multiset descends to $\mathbb{F}_q$. Before specialization, each root has multiplicity 96. Coincident values have their multiplicities added.

The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 contribution and one weight-96 contribution. Its resulting multiplicity is 144, 168, 240, or 264. No other bin has one of these multiplicities. Thus the multiset canonically recovers $-1/z$, including every modular collision.

D.2 A marginal-moment separator

For a four-party set T, let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.4),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\operatorname{rank}(L_C(T)+L_C(U)+L_C(V))}. \tag{D.2}$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12}. Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r}, where r is the $\mathbb{F}_q$-rank in (D.2).

Index a four-set by its omitted pair, an edge of K_6. The rank in (D.2) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X-lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z-lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*), \tag{D.3}$$

where b counts perfect matchings concurrent in both configurations.

Lemma D.2 (Concurrent matchings on a conic)**.** Let S be six distinct points on a nonsingular conic over a finite field k of odd characteristic other than 5. At most six perfect matchings of S have concurrent joining chords.

Proof. Identify the conic with $\mathbb{P}^1$. Projection from a point off the conic pairs the points in every fiber and hence induces a projective involution; conversely, the chords joining the three two-element orbits of a projective involution meet at its center. Thus the concurrent matchings are exactly the fixed-point-free involutions in

$$H = \mathrm{Stab}_{\mathrm{PGL}_2(k)}(S).$$

The correspondence is injective because an element of $\mathrm{PGL}_2(k)$ is determined by its action on three points. The group H acts freely on ordered triples of distinct points of S, so $|H| \mid 6 \cdot 5 \cdot 4 = 120$.

Dickson's classification, including the p-regular cases [Dic01, §§239–261], and Faber's p-irregular classification [Fab23, Theorem 1.1] are used only to obtain a uniform upper bound of six; no classification of the six-sets is needed. The possible cyclic, dihedral, A_4, S_4, and faithful degree-six A_5 cases contribute at most $1, 6, 3, 6, 0$ fixed-point-free involutions, respectively. For A_5, each involution fixes two of the six points. In the dihedral case, if the rotation order exceeded six, every orbit away from the two rotation-fixed points would have size at least that order, so no invariant six-set could exist; for rotation order at most six the two reflection classes together contribute at most six fixed-point-free elements (and the possible central half-turn contributes only when one reflection class has fixed points). In the characteristic-three semi-elementary case, the normal 3-group has order 3, while its cyclic complement embeds in $\mathrm{Aut}(C_3)$, giving only C_3 or S_3. The subfield groups whose order divides 120 are $\mathrm{PSL}_2(3) \cong A_4$ and $\mathrm{PGL}_2(3) \cong S_4$, already listed. The remaining characteristic-five subfield possibilities are excluded by the hypothesis. Thus every case is bounded by six. □

Theorem D.3 (Uniform H_3/GRS separation)**.** Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H_3 (icosahedral) sextuple

$$((0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, put $q = |k|$. Its equal-phase state is not LU-equivalent, even after a tensor-factor permutation, to any six-point GRS state over k. (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H_3 arc and its Gale dual. The other five determinant norms are -64 and -4, so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H_3 moment multiset therefore contains 70 copies of q^{-4}.

Lemma D.2 gives at most six concurrent matchings on a conic. The bound is attained whenever $\mathrm{PGL}_2(q)$ contains an octahedral S_4, by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4}, and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X - 3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H_3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (D.2) is LU-invariant, so the gap proves the claim. □

The five nonconcurrent H_3 matchings form a one-factorization of K_6, that is, a partition of its fifteen edges into five perfect matchings. The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral diagonal equivalence to the dual returns the H_3 code through an odd pentad permutation in every odd reduction. The orientation records which CSS summand is designated X and which is designated Z; local Clifford equivalence forgets this choice, so it does not define a separate local-unitary class.

D.3 The exact dimension-thirteen four-copy witness

Use the permutation contraction $I_{\boldsymbol{\sigma}}$ and matching system $M_{\boldsymbol{\sigma}}(G)$ from (6.1).

Theorem D.4 (Exact four-copy separator at $q = 13$)**.** The two regular $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after an arbitrary tensor-factor permutation.

Proof. At $q = 13$, the regular non-GRS pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\boldsymbol{\sigma})_i = \sigma_{\pi^{-1}(i)}$, and sum (6.1) over all tensor-factor permutations:

$$J_{\boldsymbol{\sigma}}(\Psi) = \sum_{\pi \in S_6} I_{\pi\boldsymbol{\sigma}}(\Psi).$$

In $\mathbb{F}_{13}$, one has $4/9 = 12$, while 4 lies on neither component of the transport divisor. Theorem E.2 therefore gives rank 21 for all 720 terms at $z = 4$, and ranks 20 and 21 with multiplicities 192 and 528 at $z = 12$. Therefore

$$J_{\boldsymbol{\sigma}}(z=4) = 720 \cdot 13^{-9}, \qquad J_{\boldsymbol{\sigma}}(z=12) = 3024 \cdot 13^{-9}. \tag{D.4}$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (D.4) makes no claim that four copies are globally minimal. □

Over $\mathbb{Q}(t)$, every fixed four-copy matching matrix has generically constant rank, so (D.4) detects a rank-jump stratum rather than a generic orbit coordinate. Section 6 gives the general distinction between these scalar contractions and the operator-valued reduced states used in the rigidity theorem.

E Transport calculation for the four-copy witness

We now explain the rank jump behind (D.4). Fix the six copy permutations

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \to k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b, $0 \le a, b < 4$, and a color-i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{\eta(i)}$ as both restriction maps for a coordinate assignment $\eta \in S_6$. This is a transport system: each edge equation requires the vertex data at its two ends to agree after applying the corresponding coordinate functional. A solution is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{\eta(i)}(x_a - y_{\sigma_i(a)}) = 0. \tag{E.1}$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.1).

Let $U = k^4/k(1,1,1,1)$, and let ρ_i be the action of σ_i on U. For each η, the MDS property makes $g_{\eta(3)}, g_{\eta(4)}, g_{\eta(5)}$ a basis of V^*. Define Q_η by

$$g_{\eta(i)} = \sum_{k=0}^{2} (Q_\eta)_{ik} g_{\eta(3+k)} \qquad (0 \le i < 3). \tag{E.2}$$

Thus Q_η is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\widetilde{Q}_\eta = 2(t-1)Q_\eta$. This scalar is a unit on the regular non-GRS locus. For the identity assignment, the polynomial representative is

$$\widetilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \qquad \begin{aligned} a &= -2(t-1)^2, & b &= -(t^2 - t + 1), \\ c &= -(t^2 - 3t + 1), & d &= -2(t-1). \end{aligned}$$

If $Y_k \in U$ are the three free copy vectors, then (E.1) reduces to the 9-by-9 block transport operator

$$\widetilde{T}_\eta = \big((\widetilde{Q}_\eta)_{ik}(\rho_i - \rho_{3+k})\big)_{i,k=0}^{2}. \tag{E.3}$$

The dimension identity

$$21 - \operatorname{rank} M_{\pi\boldsymbol{\sigma}}(G(t)) = 9 - \operatorname{rank} \widetilde{T}_\eta \tag{E.4}$$

holds over the field k, where $\eta = \pi$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the regular non-GRS six-arc locus.

The remaining calculation has an internal group symmetry. The next lemma reduces the 720 coordinate assignments to four double cosets for each rank-drop question. Cycles act on the coordinate labels $1, \ldots, 6$.

Lemma E.1 (Double-coset reduction)**.** Define

$$H_s = \langle (34)(56), (3546), (12)(56), (13)(24)(56) \rangle, \quad K_s = \langle (36), (23)(46), (12)(45) \rangle,$$
$$H_a = \langle (56), (34), (35)(46), (12), (13)(24) \rangle, \qquad K_a = \langle (36), (23)(46), (15) \rangle.$$

Then $(|H_s|, |K_s|) = (24, 48)$ and $(|H_a|, |K_a|) = (48, 16)$. The signed double cosets

$$H_s \backslash S_6 / K_s \quad \text{have representatives} \quad 1, (56), (456), (235)$$

have sizes $192, 192, 288, 48$, respectively. The axial double cosets

$$H_a \backslash S_6 / K_a \quad \text{have representatives} \quad 1, (456), (2345), (235)$$

have sizes $384, 192, 96, 48$, respectively.

At a generic point of $3B - 2A = 0$, the reduced transport operator has rank eight exactly on $H_s K_s$. At a generic point of $3B + 2A = 0$, it has rank eight exactly on $H_s(56)K_s$. At a generic point of $B^2 - 2A^2 = 0$, it has rank eight exactly on $H_a(2345)K_a$. On all other double cosets its rank is nine.

Proof. The displayed generators act on $\widetilde{T}_\eta$ by invertible row and column operations on the indicated divisor, so rank is constant on the corresponding double cosets. Multiplying the generators gives subgroup orders $24, 48, 48, 16$. Their intersections at the listed representatives have orders $6, 6, 4, 24$ in the signed case and $2, 4, 8, 16$ in the axial case. The formula

$$|HrK| = |H||K| / |H \cap rKr^{-1}|$$

therefore gives the displayed sizes; in each case the four sizes sum to $|S_6| = 720$, so the lists are exhaustive.

It remains to evaluate one 9-by-9 operator per double coset. Using $a = b + c$, $d^2 = -2a$, $A = a(c - b)$, and $B = bc$, fraction-free elimination gives, at the three rank-dropping representatives,

$$\begin{aligned} \det \widetilde{T}_1 &= 64 d^3 AB(3B - 2A), \\ \det \widetilde{T}_{(56)} &= 64 d^3 AB(3B + 2A), \\ \det \widetilde{T}_{(2345)} &= 64 d^3 A(B^2 - 2A^2). \end{aligned}$$

The representative determinants on the other double cosets are nonzero modulo the corresponding factor. A nonzero 8-minor remains on each of the three components, so every stated rank drop is exactly from nine to eight. □

Theorem E.2 (Transport rank-drop locus)**.** For a parameter on the regular odd non-GRS pencil, there exists a coordinate assignment $\eta \in S_6$ for which the transport system (E.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z - 2)(9z - 4) = 0. \tag{E.5}$$

At a generic point above $z = 2$, exactly 96 coordinate assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. Lemma E.1 shows that the only generic rank-drop components are $3B \pm 2A = 0$ and $B^2 - 2A^2 = 0$. The first two are the two sheets $B/A = \mp 2/3$ above $z = 4/9$, and each supports 192 coordinate assignments. The last is $z = 2$ and supports 96 assignments. The nonzero 8-minors make every reduced kernel one-dimensional. Equation (E.4) turns these kernels into rank 20 of the matching matrix and gives rank 21 everywhere else. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.2.

Corollary E.3 (Exceptional characteristics)**.** On the regular locus, characteristic 3 has no rank-drop component. In characteristic 5, only the $z = 4/9$ component disappears, because it is GRS. In characteristic 7, the two components merge; exactly 288 assignments then have rank 20, and the remaining 432 have rank 21. In every other odd characteristic the two components and their generic multiplicities remain as in Theorem E.2.

Proof. The identity $4B^2 + A^2 = 4G^2$ specializes to

$$\begin{aligned} \operatorname{char} k = 3: &\quad B^2 - 2A^2 = G^2, \qquad 9B^2 - 4A^2 = -A^2; \\ \operatorname{char} k = 5: &\quad 9B^2 - 4A^2 = 4G^2; \\ \operatorname{char} k = 7: &\quad 9B^2 - 4A^2 = 2(B^2 - 2A^2). \end{aligned}$$

The regular-locus exclusions give the first two claims. In characteristic 7, the disjoint 96- and 192-element double cosets of Lemma E.1 give the stated histogram. The only other primes in the resultants of the representative determinants with the excluded boundary factors are $11, 13, 41$; direct reduction of the displayed nonzero 8-minors shows that these cause ramification but no further rank drop. $\square$

The paper-local certificate independently replays the representative determinants and their reductions.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.1, cannot be promoted to a generic coordinate.

F Coordinate-permutation extensions

This optional appendix records the splitting obstruction for the projective coordinate-permutation extension, together with finite examples. It may be skipped without affecting the fixed-coordinate transversal-group theorem.

The nonabelian factor set

Let Γ be the fixed-coordinate projective product-unitary symmetry group, let $\widetilde{\Gamma}$ be the group allowing coordinate permutations, and let Π be the realized permutation group, so that $1 \to \Gamma \to \widetilde{\Gamma} \to \Pi \to 1$ is exact. Operationally, splitting means that the realized coordinate permutations can be represented by product symmetries forming an actual subgroup, rather than only up to fixed-coordinate symmetries.

The extension has a section-free outer action $\Pi \to \mathrm{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ, and changing the lift changes that automorphism by an inner automorphism, because elements of the kernel act by inner automorphisms. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi, \rho)s(\pi\rho), \qquad f_s(\pi, \rho) \in \Gamma.$$

Then $f_s(1, \pi) = f_s(\pi, 1) = 1$, and, with α_π denoting conjugation by $s(\pi)$, associativity of three chosen lifts gives

$$f_s(\pi, \rho) f_s(\pi\rho, \upsilon) = \alpha_\pi(f_s(\rho, \upsilon))\, f_s(\pi, \rho\upsilon).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$ with $b(1) = 1$, then substituting and commuting no kernel factors gives

$$f_{s'}(\pi, \rho) = b(\pi)\alpha_\pi(b(\rho))\, f_s(\pi, \rho)b(\pi\rho)^{-1}.$$

The order of the factors is essential because Γ need not be abelian. The factor set is identically one precisely when the section preserves multiplication, so the extension splits exactly when f_s can be changed to the constant identity factor set by such a normalized b.

A uniform complement for the icosahedral family

The repeated H_3 rows are reductions of one integral construction, so they admit a theorem rather than a field-by-field census.

Theorem F.1 (Uniform H_3 coordinate-permutation extension)**.** Let k be an odd prime field containing τ with $\tau^2 = \tau+1$, and suppose the reduction of the integral sextuple in Theorem D.3 is a non-GRS six-arc. Its realized coordinate-permutation image is the exceptional degree-six action of S_5, and the projective coordinate-permutation extension splits. Even coordinate permutations centralize the fixed split torus T, while odd ones invert it.

Proof. Put $\bar\tau = 1 - \tau$ and

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & \bar\tau \\ \tau & 0 \end{pmatrix}, \qquad L = \begin{pmatrix} 0 & \tau - 2 \\ \tau + 1 & 0 \end{pmatrix}.$$

The identities $\tau\bar\tau = -1$ and $(\tau - 2)(\tau + 1) = -1$ put all three matrices in $\mathrm{SL}_2(k)$. In the semidirect-product convention of this appendix, consider

$$\widehat{x} = ((34)(56); -I, -I, I, I, I, I), \qquad \widehat{y} = ((1\,2\,3\,4\,6\,5); J, J, J, K, K, L).$$

Substitution in the four CSS block equations of Section 3 shows that both elements preserve the label space. This calculation uses only $\tau^2 = \tau + 1$, so it holds before reduction over $\mathbb{Z}[\tau]$.

Let x, y be the two displayed coordinate permutations, and set

$$u = xyxy^{-1}, \qquad v = y^{-1}xy^2.$$

The elements u, v fix coordinate 1, have orders $5, 4$, and satisfy $v^{-1}uv = u^3$. They therefore generate the Frobenius group $C_5 \rtimes C_4$ of order 20. Since y is transitive, while the one-factorization recovered by the marginal incidence has stabilizer S_5 of order 120, the permutations x, y generate exactly that S_5.

The same relations hold for the lifts $\widehat{u} = \widehat{x}\widehat{y}\widehat{x}\widehat{y}^{-1}$ and $\widehat{v} = \widehat{y}^{-1}\widehat{x}\widehat{y}^2$: direct multiplication over $\mathbb{Z}[\tau]/(\tau^2 - \tau - 1)$ gives

$$\widehat{x}^2 = \widehat{y}^6 = \widehat{u}^5 = \widehat{v}^4 = 1, \qquad \widehat{v}^{-1}\widehat{u}\widehat{v} = \widehat{u}^3.$$

Schreier rewriting with transversal $1, \widehat{y}, \ldots, \widehat{y}^5$ reduces the stabilizer of coordinate 1 to $\langle \widehat{u}, \widehat{v} \rangle$. Hence the generated lift group has at most $6 \cdot 20 = 120$ elements. Its coordinate projection is the 120-element group just found, so projection is an isomorphism. This gives the required label-space complement, and the phase-correction lemma lifts it to projective Clifford symmetries. Finally, diagonal local blocks centralize T, whereas antidiagonal blocks invert it; for the displayed generators this distinction is exactly coordinate-permutation parity. □

The supplement records exact finite computations for six additional prime-field examples. They illustrate the factor-set criterion but are not used in any theorem of the paper. No classification of arbitrary coordinate-permutation extensions is claimed.

G Verification and scope

This appendix audits proof dependencies; it is not needed to follow the all-length classification. The conductor lemma and group dichotomy are proved in the text without finite enumeration. Exactness uses the imported transversal Clifford no-go, and the construction uses the imported stabilizer-character correction [Rudd26a].

Claim family	Status	Verification method	Imported or trusted input
Conductor dimension	proved here	MDS rank functions and a vanishing-coordinate argument	standard product-code interpretation
Exact fixed-coordinate group	proved here	block preservation, explicit unipotents, and logical Paulis	imported product-unitary rigidity and phase correction
Six-point conic criterion	proved here	MDS–CSS dictionary and Gale-transform argument	none
Pencil quotient	proved here	symbolic bracket identities and a ten-orbit holonomy formula	imported minimum-support transition theorem; independent certificate replay
Frobenius-sector formulas	proved here	direct finite-field algebra	no complete extension-field orbit classification
Six-point syndrome example	consequence	syndrome lemma composed with a cited deep-hole theorem	cited conic, count, and orbit result
Scalar separators and transport divisor	proved here	concurrence bounds, double-coset reduction, and representative determinants	independent certificate replay
Coordinate-permutation extensions	one family proved	integral S_5 complement	supplementary finite examples; no general classification
Formalization	partial	Lean kernel checking	hypothesis-explicit interfaces and three trusted finite cardinality evaluations

Table 1: Proof and verification status of the main claim families. A hypothesis-explicit interface proves a conclusion from named mathematical or computational inputs; importing it does not prove those inputs.

The paper-local supplement contains the exact generators, compact JSON certificates, and a deterministic standard-library verifier for the six-point computations. The verifier checks the manifest and regenerates every certificate. For each finite claim, it either checks a compact witness directly or reruns the stated exhaustive or symbolic computation and compares the canonical output. The supplement identifies which route is used and records each finite domain, stopping condition, independent replay, and negative scope statement. Repository documentation gives the exact replay command.

The Lean development formalizes the MDS–CSS dictionary, conductor dimension, pencil quotient, extension-field formulas, syndrome geometry, contraction and transport algebra, and abstract consequences of a supplied coordinate-permutation complement. The finite classifications and explicit separators remain certificate inputs to hypothesis-explicit interfaces. Three finite graph cardinalities are trusted exhaustive evaluations, and two marginal-moment results inherit one of those assumptions. Repository documentation records the exact formal entry points, toolchain, and dependency audit.

The local-unitary classification by z is restricted to odd prime fields. Over extension fields, Frobenius is already an additional local Clifford and the full additive-Clifford orbit problem is not claimed. The exact group theorem fixes the physical coordinates and assumes a product implementation. It does not classify arbitrary coordinate-permutation enlargements, non-product physical implementations, nonstabilizer AME states, or unrelated quantum-code families.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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